Priors: $P(yes) = \frac{7}{15}$, $P(no) = \frac{8}{15}$

Conditional Probabilities:

laplace smoothing

$P(square \mid yes) = \frac{3+1}{7+2} = \frac{4}{9}$

$P(large \mid yes) = \frac{6+1}{7+2} = \frac{7}{9}$

$P(red \mid yes) = \frac{3+1}{7+3} = \frac{4}{10} = \frac{2}{5}$

laplace smoothing

$P(square \mid no) = \frac{5+1}{8+2} = \frac{6}{10} = \frac{3}{5}$

$P(large \mid no) = \frac{2+1}{8+2} = \frac{3}{10}$

$P(red \mid no) = \frac{3+1}{8+3} = \frac{4}{11}$

Bayes' Theorem

$$P(yes \mid square, large, red) \overset{\downarrow}{=} \frac{P(square, large, red \mid yes)\, P(yes)}{P(square, large, red)}$$

Naive Bayes assumes feature are independent

$$= \frac{P(square \mid yes)\, P(large \mid yes)\, P(red \mid yes)\, P(yes)}{P(square, large, red)}$$

$$= \frac{\frac{4}{9} \cdot \frac{7}{9} \cdot \frac{2}{5} \cdot \frac{7}{15}}{0.0645 + 0.0349} = \frac{0.0645}{0.0994} = \boxed{0.649}$$

Bayes' Theorem

$$P(no \mid square, large, red) \overset{\downarrow}{=} \frac{P(square, large, red \mid no)\, P(no)}{P(square, large, red)}$$

Naive Bayes assumes feature are independent

$$= \frac{P(square \mid no)\, P(large \mid no)\, P(red \mid no)\, P(no)}{P(square, large, red)}$$

$$= \frac{\frac{3}{5} \cdot \frac{3}{10} \cdot \frac{4}{11} \cdot \frac{8}{15}}{0.0994} = \frac{0.0349}{0.0994} = \boxed{0.351}$$